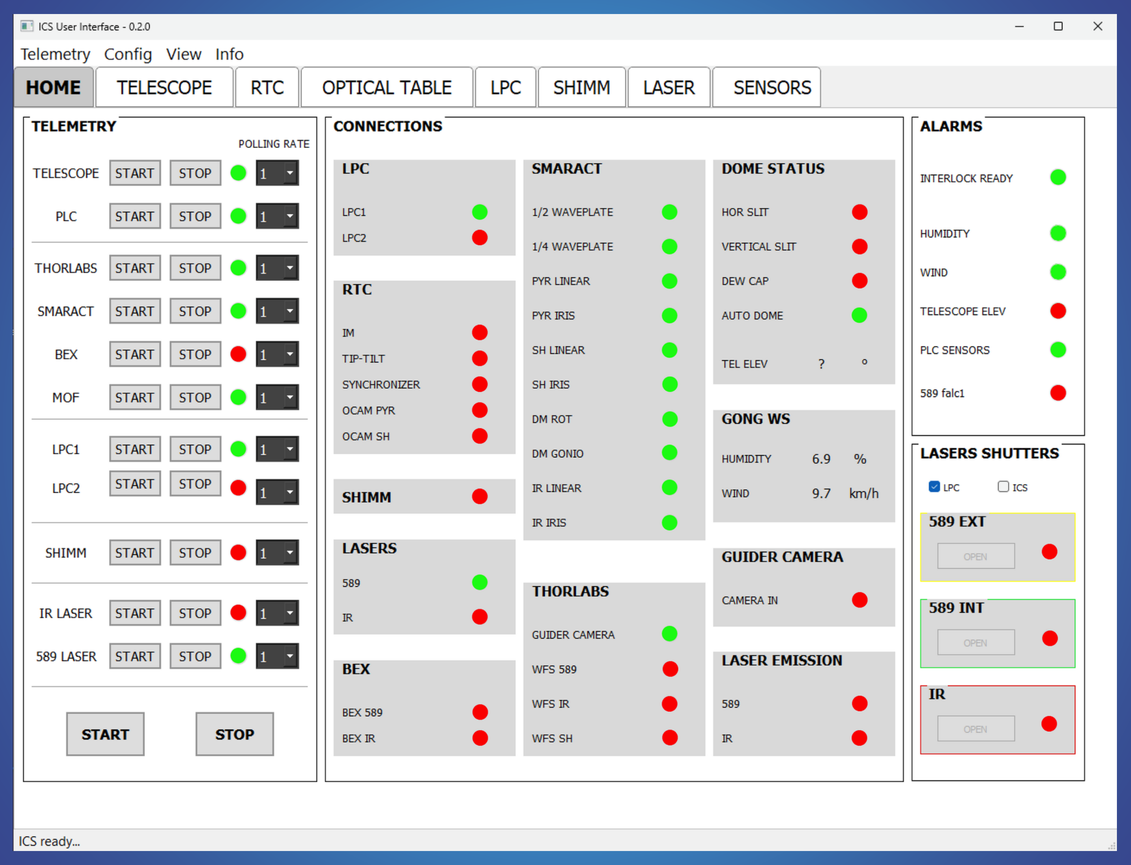
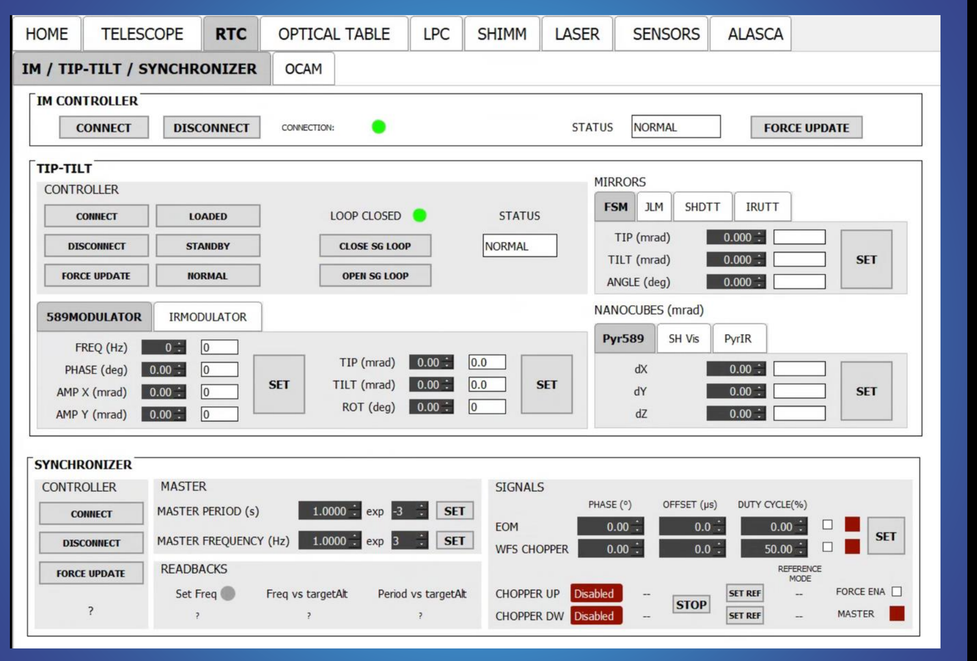
A Python-Based Instrument Control System for Adaptive Optics: Architecture and Applications in the CaNaPy–ALASCA Experiment

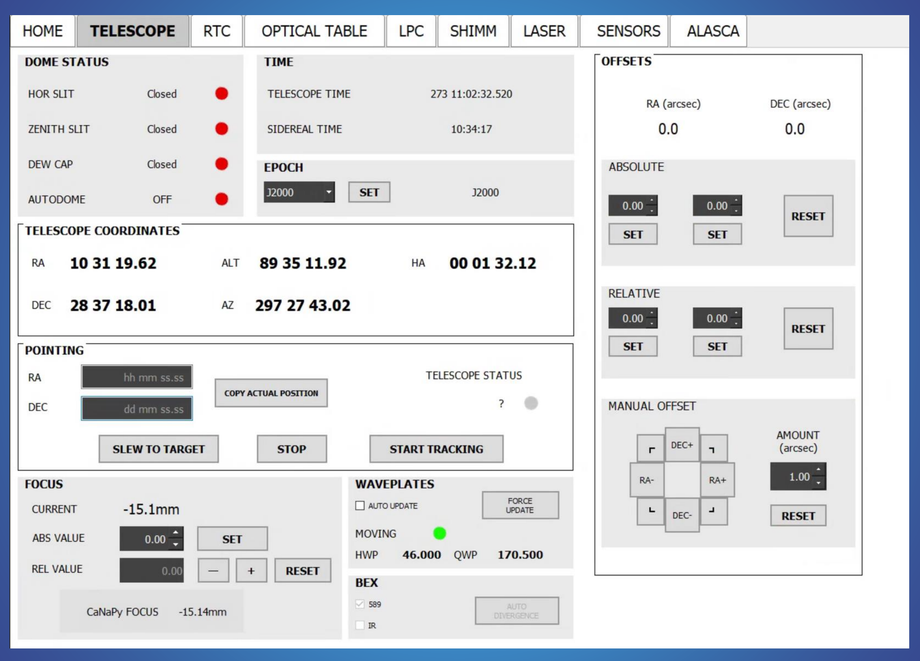
Lorenzo Paonni, Ivan Agostinelli, Mauro Centrone, Marco Faccini, David Jenkins, Felipe Pedreros Bustos, Dietrich Pescoller, Roberto Biasi, and Laura Salvi
lorenzo.paonni@inaf.it

INTRODUCTION

The Instrument Control System (ICS) is a Python-based software framework developed to manage and coordinate the diverse opto-mechanical and electro-optical devices used in advanced adaptive optics (AO) facilities. Modern AO experiments require a high level of automation and reliability, with seamless integration of telescopes, lasers, cameras, and wavefront sensors into a unified environment. The ICS addresses this need by providing a modular and extensible platform that simplifies configuration, operation, and monitoring of complex hardware setups. Developed within the CaNaPy–ALASCA collaboration between ESO, ESA, and partner institutions, the ICS plays a central role in enabling innovative AO experiments such as laser guide star uplink pre-compensation and satellite communication demonstrations, bridging laboratory validation and on-sky operation.





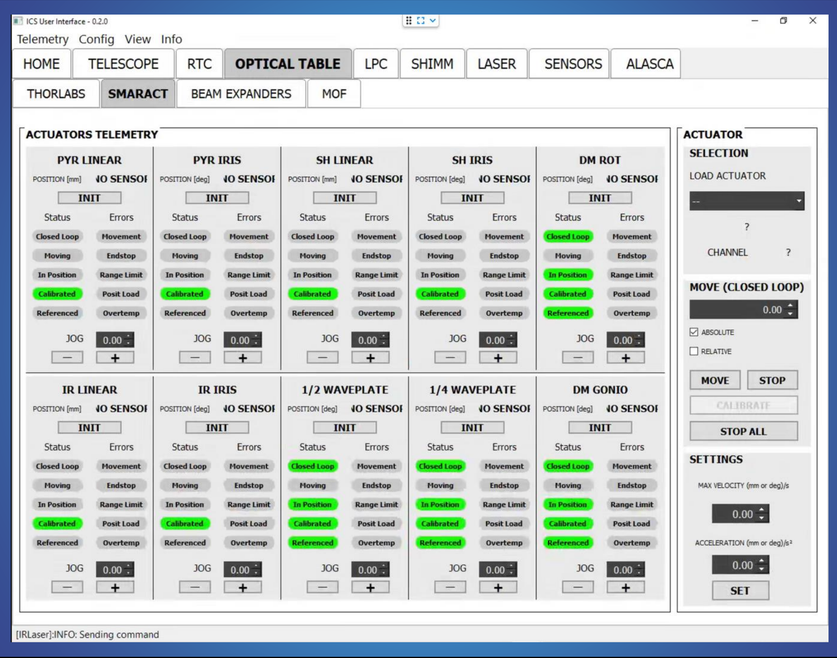


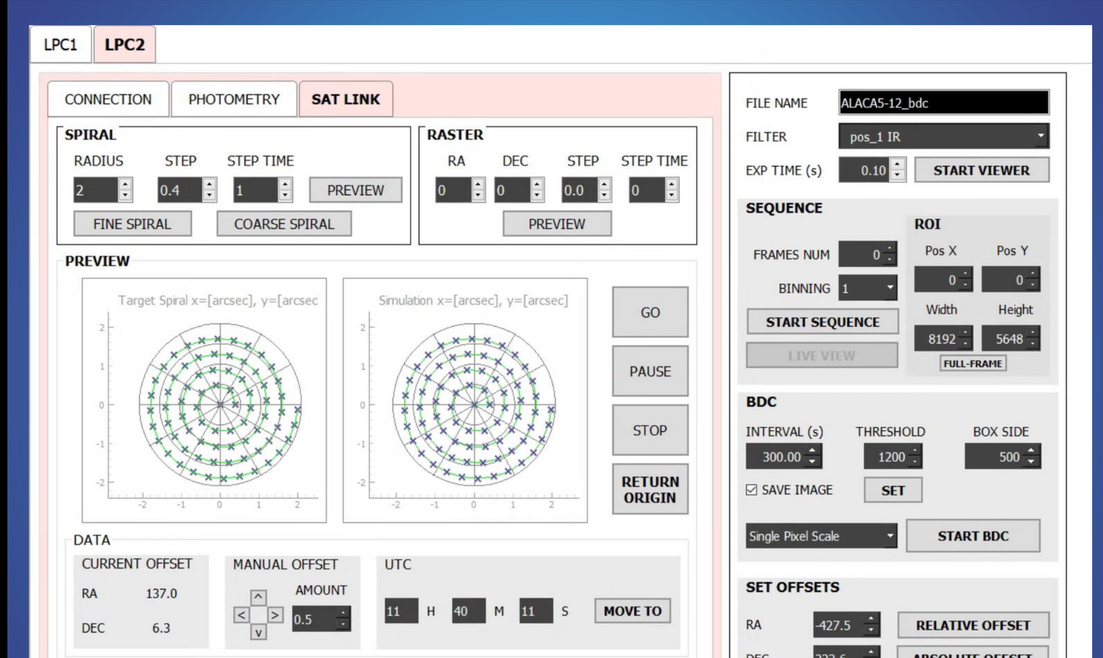
APPLICATIONS

Within the CaNaPy project, the ICS plays a central role in demonstrating laser guide star (LGS) uplink pre-compensation and wavefront sensing with a Pyramid WFS. The system supports investigations into atmospheric tip-tilt retrieval using the time-delay method. In the ALASCA experiment, the ICS extends its scope to satellite optical feeder links, implementing uplink pre-compensation to ensure stable ground-to-space laser communication under real atmospheric conditions.

SYSTEM OVERVIEW

The ICS integrates multiple subsystems, including the Optical Ground Station telescope, laser pointing cameras, 589 nm and infrared lasers, beam expanders, and motorised stages. It ensures synchronisation with external systems such as the Real-Time Controller (RTC), Telescope Control System (TCS), and laser shutters. A dedicated graphical user interface supports both manual and automated workflows, allowing repeatable and parameter-driven experiment execution. Comprehensive telemetry and logging capabilities enable performance tracking and facilitate troubleshooting.





RESULTS

The ICS has been successfully deployed during on-sky and laboratory campaigns, where it has proven its robustness in coordinating complex hardware environments. The integration with the RTC has enabled dynamic adaptive loop closure, while reliable synchronisation with telescope and laser systems has ensured smooth operation during critical experiments. Logging of environmental and configuration parameters has been key for performance evaluation, supporting iterative improvement of AO control strategies.

CONCLUSIONS

The Python-based ICS provides a flexible, reliable, and extensible solution for adaptive optics experiments requiring precise control of diverse hardware subsystems. Its successful application in the CaNaPy–ALASCA facility highlights its capacity to bridge laboratory testing and operational on-sky demonstrations. As AO systems continue to evolve, the ICS framework offers a strong foundation for future developments in both astronomical and satellite communication contexts.